

Optical power budget calculating tool for fibre in the access network

P Bell, K Cobb and J Peacock

This paper details the importance of network modelling to aid the process of network design and planning. In particular, the paper describes the use of statistical network modelling tools in determining the optical power budget of PON and point-to-point topologies. The methodology adopted by BT is described together with the issues which need to be considered when developing such network modelling tools. The paper then outlines the development and implementation of a computer-based planning tool that is currently being implemented within BT. Finally, predictions from the tool are compared with actual deployed network measurements in order to establish its overall accuracy.

1. Introduction

The deployment of optical fibre into the access network has highlighted several issues which need to be addressed. These are namely: 'How is the optical performance of any given network design predicted?' and: 'How during actual deployment is this performance then guaranteed?' Valuable experience has been gained from a variety of field trials with regard to practicalities of deployment of optical fibre [1, 2]. This experience, together with research into the use of computer aided design (CAD), has identified the importance of computer-based network design and planning tools [3], especially for emerging passive optical networks (PONs).

One of the core requirements of any computer-based planning system is the ability to calculate the optical forward and return losses, the power budget, and associated system margin [3]. This paper considers issues concerned with the power budget and the ability to predict simply, but accurately, the optical power loss associated with planned architectures. The paper also discusses aspects specific to the implementation of this tool and details comparisons between predictions from the tool and actual deployed networks.

2. The importance of power budget planning

Accurate prediction of the optical power losses, associated with proposed networks, is fundamental to a successful fibre deployment strategy.

The introduction of passive optical network architectures has increased the importance of predicting accurately a deployed network's power budget. The 32-way PON architecture recently adopted by BT for use in the access

network can reduce deployment costs dramatically compared to point-to-point networks [4, 5]. However, the losses associated with the splitters can significantly encroach upon the maximum allowable power loss, and in general PONs will operate with smaller operating margins than historically deployed point-to-point fibre links.

A second reason for requiring a precise method of network optical power loss prediction, is the relationship between the network power loss and the system reliability. For example, it can be shown that the effect of transient attenuation upon the transmission performance of an optical fibre network is inextricably linked to the power loss and the associated system margin of that network [6, 7]. Thus, knowledge of the power loss associated with an optical fibre network design is essential to understanding its in-service performance and reliability.

Finally, a method of determining accurately the optical loss at a specific location in a deployed network, allows a simplified network evolution path. This is imperative if a stable and 'future-proof' optical-fibre access network is to be achieved.

3. Modelling approaches

Historically, the calculation of the power loss associated with a particular optical fibre network was based upon a worst-case approach, i.e. the maximum loss of each component. The total loss predicted for a proposed network architecture was then based upon the addition of all the maximum losses of the components. The maximum loss method is inappropriate in the access network (particularly

for PONs), because of the larger number of passive components, and smaller margins, which pertain to any transmission path through the network.

Thus, a more sophisticated methodology is required. A more powerful approach is to use a statistical analysis, taking account of the statistical loss distribution of each component in the network. This then allows for components with lower and higher than expected losses. A statistical approach also enables the use of risk management in the deployment of PON topologies. Furthermore, the method can be used to determine the confidence with which the optical power loss of a particular network is predicted. This could complement the decision about whether to use PON or point-to-point architectures.

4. Statistical modelling methodology

The method employed by BT to calculate an optical link power budget is the subject of an ETSI standard [8, 9]. This method has evolved from the observation that sample losses of optical components (e.g. splitters, fibre, splices and connectors) vary approximately according to a normal distribution with appropriate mean and standard deviation.

Under this assumption, the end-to-end power loss associated with any fibre link may be represented statistically using the properties of normal distributions.

Utilising the convolution of normal distributions, consider a network consisting of n different components with means $m_1, m_2, \dots, m_n$, and standard deviations $\sigma_1, \sigma_2, \dots, \sigma_n$. If each component occurs $a_1, a_2, \dots, a_n$ times, then the total mean m_T and total standard deviation σ_T are given respectively by

$$m_T = \sum_{i=1}^{i=n} a_i m_i \quad \text{..... (1)}$$

and:

$$\sigma_T^2 = \sum_{i=1}^{i=n} a_i \sigma_i^2 \quad \text{..... (2)}$$

It should be noted that the total standard deviation is given by the square-root of the sum of the squares, rather than just the sum. Furthermore, care must be taken when considering cable to be one or more of the n components. The mean loss of any cable component should be expressed in dB per unit length, and the associated standard deviation should be expressed in dB per square-root of unit length, while the number of occurrences should be expressed in number of unit lengths. In the analysis detailed later the unit

length is one metre. The losses associated with lengths of less than one metre may be determined by applying equations 1 and 2, but would be insignificant.

Once the total mean loss and the associated standard deviation have been determined it is relatively simple to determine 'confidence' intervals [10]. These intervals are constructed using the total mean and total standard deviation. Table 1 details three possible confidence intervals that may be determined.

Table 1 Confidence intervals associated with a normal distribution.

Confidence interval	Minimum loss	Maximum loss
68.3%	$m_T - \sigma_T$	$m_T + \sigma_T$
95.5%	$m_T - 2\sigma_T$	$m_T + 2\sigma_T$
99.7%	$m_T - 3\sigma_T$	$m_T + 3\sigma_T$

5. Fibre infrastructure deployment concerns

In applying the above method during the planning process of optical fibre deployment, account must be taken of the intended service lifetime of the network. Hence, several factors other than the network components must be catered for. The main factors are a maintenance allowance and the use of supervisory tools such as a clip-on power meter.

An allowance for maintenance (in terms of re-routing perhaps) will ensure the longevity of the network. For example, a successful network strategy must anticipate a degree of cable 'dig-up'. Failure to do so would render part of or a whole network useless in the event of a catastrophic breakage.

Catering for the use of supervisory tools such as a clip-on power meter [11] will enable monitoring of an optical fibre network (using wavelength division multiplexing) without degradation of service. This aids fault diagnostics and leads to improved quality of service.

Thus, a successful network power budget calculation must take account of not only the network component losses, but also the additional lifetime factors.

6. Power budget tool implementation

Of importance with the implementation of any calculating planning tool is the platform upon which the tool will be designed. The choice of platform will determine the availability of the tool to network planners. In order to guarantee convenience of use, especially with initial development and trial of the tool, a platform which is readily available to fibre planners should be sought. For this reason, an algorithm based upon the statistical methodology and taking account of the additional lifetime factors, was written using Microsoft® Excel.

The tool was developed in three distinct modules in order to maintain flexibility. These modules may be defined as the parameters module, the manual calculation module and the automation module, as shown in Fig 1.

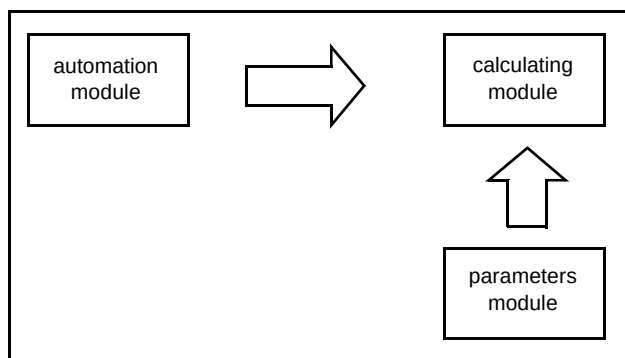


Fig 1 A schematic of the power budget calculating tool.

The parameters spreadsheet details the optical power loss data of all the components associated with an optical fibre network, together with the allowances for the lifetime factors (see section 5). These variables are stored as a separate module in the tool so that updating the data is a simple procedure.

The automation module provides the planner with a simple user interface. The planner is guided through a generic optical fibre architecture, allowing for either PON or point-to-point topologies. Wherever possible the tool has been made as general as possible in order to encompass as many network scenarios as possible. Once data about the proposed network architecture has been inserted, the automation module then manipulates the calculating module to apply the user's scenario. The output to the user is the calculated mean and confidence interval of the optical power loss associated with the proposed network.

Finally, the tool has been designed so that the experienced user may also run the model manually by operating the calculating module. This gives a greater degree of flexibility and provides more power budget information.

7. Comparison of results

Initial results from the tool have been compared with two actual reference network topologies — a point-to-point and PON network. These reference networks were deployed using OTIAN[®] optical plant [12–15], with the PON constructed with a 32-way split. The attributes of the two net-

works are detailed in Table 2. Typical values of the component mean and standard deviation losses [5, 8] for both transmission windows are shown in Table 3.

Table 2 The components of the point-to-point optical link and the PON considered.

Component	Point to point	PON
Fibre length, m	3210	3560
Fusion splices	8	10
2 × 4 splitter	—	1
2 × 8 splitter	—	1

Table 3 Typical component mean and standard deviation (SD) optical power losses used.

Component	1300 nm window		1500 nm window	
	Mean loss (dB)	SD loss (dB)	Mean loss (dB)	SD loss (dB)
Fibre/m	0.4	0.05 ¹	0.3	0.05 ¹
Fusion splicers	0.2	0.03	0.2	0.03
2 × 4 splitter	6.7	0.45	6.7	0.45
2 × 8 splitter	9.8	0.57	9.8	0.57

Figures 2 and 3 illustrate the spectral loss characteristic of both these networks. The estimated 99.7% confidence interval of the loss (in both transmission windows) is also shown (using component mean and standard deviation losses similar to the draft ETSI standard [8]). An upper bound (99.7% confidence) for the allowed additional lifetime losses is included in both transmission windows.

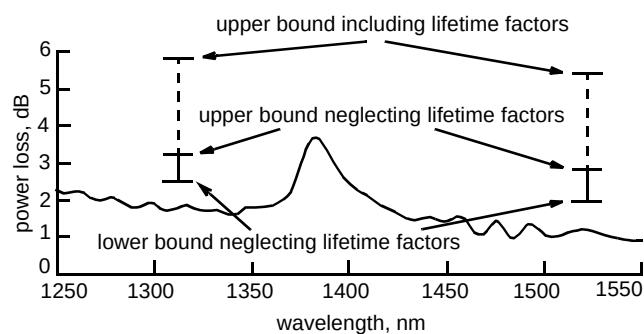


Fig 2 Point-to-point network (see Table 2) estimated (99.7% confidence) and measured optical power losses. Also included is the upper bound including lifetime factors.

¹ Note that the units are dB/m ^{1/2}.

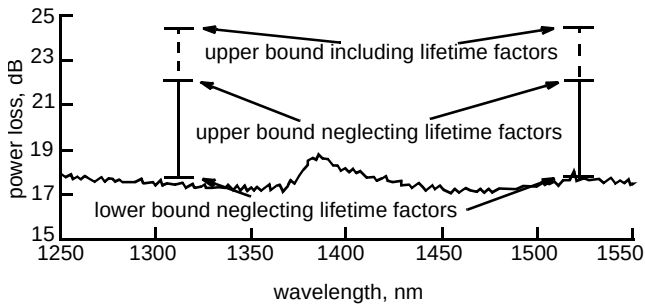


Fig 3 PON network (see Table 2) estimated (99.7% confidence) and measured optical power losses. Also included is the upper bound including lifetime factors.

Figures 2 and 3 illustrate that the measured spectral loss was consistently lower than the minimum computed value. This anomaly may have several explanations. Firstly, the assumption that the loss associated with the different network components follows Gaussian normal statistics may be an inappropriate approximation. Another explanation may lie in the fact that the confidence interval only applies to 99.7% of all occurrences; the figures shown may in fact lie in the 0.3% region.

A more probable explanation is that the values of the mean and standard deviation losses (see Table 3) are inaccurate, i.e. the losses associated with the components are pessimistic, leading to a confidence interval above the measured spectral losses. It should be noted that the fact that the calculated losses were pessimistic, may be indicative of a cautious approach. Further refinement of the mean and standard deviation losses may provide more consistent results. Furthermore, a pessimistic approach would guarantee service in the main, with only extreme cases of receiver saturation causing difficulties.

Several other results should also be noted from Figs 2 and 3. Firstly, the upper bound including the lifetime factors (see section 5) can have a significant impact upon the power budget. Figures 2 and 3 show an additional 3 dB due to the lifetime factors, although it should be noted that the actual allowance will, to some extent, be dictated by the level of acceptable risk.

Furthermore, the size of the confidence interval for the PON network is far greater than that for the point-to-point network. The reason for this is twofold; there are more network components in the PON network, but, more importantly, the introduction of the fusion splitters has increased the variability across the network. This is shown

in Figs 2 and 3 which show the confidence intervals of approximately 1 dB and 4 dB, for the point-to-point and PON networks respectively.

Finally, it should be pointed out that the losses associated with the PON network are significantly greater than the point-to-point network. This reiterates the point raised in section 2, that the losses associated with the splitters can significantly encroach upon the maximum allowable power loss.

8. Conclusions

This paper has shown the importance of determining accurately the power loss associated with any proposed optical fibre access network. The relative merits of the use of a statistical approach to estimating the loss have been considered. The theoretical basis for this statistical approach to network power loss modelling has been detailed.

Furthermore, the need to consider the network's lifetime (specifically, maintenance and clip-on power budget allowance) and its effect upon the maximum allowable power loss has also been highlighted.

The tool developed in the Excel environment has been described and preliminary results investigated. The results indicate that while initial values used in the tool over-estimate the end-to-end optical power loss (see Figs 2 and 3), the factors for this are well understood and further analysis will refine the tool's accuracy. The tool and methodology represent a significant improvement upon previous methods and is currently being utilised by BT's fibre planners.

Acknowledgements

The authors would like to thank Tim Finegan (BT) for all his help and advice, and James McGowan is also acknowledged for his efforts in developing the statistical tool.

References

- 1 Ritchie W K: 'Overview of local access development', BT Technol J, 7, No 2, pp 7—16 (1989).
- 2 Hoppitt C E, Astbury M L, Keeble P and Chapman P A: 'Operations and maintenance experience of the Bishop's Stortford fibre trial', 3rd IEE Conference on Telecommunications, No 331, pp 165—168 (1991).
- 3 Asumu D E and Senior J M: 'Software package for designing/planning networks in the subscriber loop', Fibre and Integrated Optics J, 15, No 1 (January 1996).
- 4 Dalgoutte D G and Lea-Wilson N D: 'Installation cost issues in external plant', 13th Annual Conference on European Fibre Optic Communications and Networks proceedings, 3, (June 1995).

- 5 Mellis J, Rabone N H, Redstall R M and Shaw R: 'Performance and reliability requirements for passive components in local optical networks', 12th Annual Conference on European Fibre Optic Communications and Networks proceedings, 3 (June 1994).
- 6 Peacock J: 'Measurements of optical transient attenuation losses and their effects on deployed optical systems in the British Telecom network', Electronics Letters, 30, No 20 (September 1994).
- 7 Franckx J, Peacock J, Cobb K and Bell P: 'Investigations into the optical performance of a new generation of network plant items', 13th Annual Conference on European Fibre Optic Communications and Networks proceedings, 3, (June 1995).
- 8 'Transmission and Multiplexing (TM): Optical distribution network for optical access networks (OANs)', ETSI Draft (DE/TM-01009) (February 1995).
- 9 Bell P, Cobb K and Peacock J: 'Improved interval estimation of fibre loss budgets', 2nd Communication Symposium, Manchester Metropolitan University, pp 45—49 (July 1995).
- 10 Hambourg M: 'Basic statistics', 2nd Edition, Harcourt, Brace and Jovanovich (1979).
- 11 Ferguson D A and Mellis J: 'New fault diagnostics techniques for PONs in the access network', 13th Annual Conference for European Fibre Optic Communications and Networks proceedings, 3 (June 1995).
- 12 Brewer D A, Hale P G, Peacock J and Walling R: 'Modular optical plant for access network: operational aspects', 13th Annual Conference on European Fibre Optic Communications and Networks proceedings, 3 (June 1995).
- 13 Hale P G, Peacock J and Bell P: 'Modular optical plant for the access network: a practical solution', 13th Annual Conference on European Fibre Optic Communications and Networks proceedings, 3 (June 1995).
- 14 Dalgoutte D, Hewins A, Hornung S and Peacock J: 'OTIAN — an outside plant system for the access network', 13th Annual Conference on European Fibre Optic Communications and Networks proceedings, 3 (June 1995).
- 15 Frost P and Peacock J: 'OTIAN — an infrastructure system for networks in the future', Communications Symposium, Manchester Metropolitan University, pp 97—100 (July 1994).



Peter Bell graduated from Coventry Polytechnic in 1990 obtaining a first class honours degree in Mathematics. In 1993, he graduated from Newcastle University with a PhD in Mathematics.

Since joining BT Laboratories in 1993, he has worked in the external plants and pilots unit, within the network transport engineering centre. This work has involved analysing the performance of fibre and copper in the access network. In particular, the work has aided the development of fibre deployment planning rules.

He is a member of the Institute of Mathematics and its Applications, and the Institute of Electrical Engineers.



John Peacock joined BT in 1980 in Leeds after spending 5 years in the communications branch of the Royal Navy. In 1983 he attended Leeds Polytechnic and graduated in 1987 with a first class honours in Communications Engineering. He then joined BT Laboratories and has worked on a number of access network related projects, including leading the teams developing the TPN specifications, OTIAN® plant system, access network cost and performance models, and access network planning rules. He currently leads a team which specialises in access network infrastructure performance and planning issues, including the development of advanced planning tools. He holds an MBA and is a Chartered Engineer and a Member of the IEE.

He holds an MBA and is a Chartered Engineer and a Member of the IEE.



Ken Cobb works in the external plant and pilots development unit at BT Laboratories working on projects addressing copper planning issues with the aim of improving current processes.

Before joining BT he worked on electrical standards at AVO followed by a BEng student apprenticeship and work on microwave communications at GEC-Marconi Communications Ltd. He joined BT in 1981 after completing an MSc at University College, London. He trialled the first multimode and single-mode standard optical fibre systems and set up mobile and static fibre measurement laboratories. He moved to BT Laboratories to head the development of in-building fibre systems using blown fibre as the main installation technique. Following the successful roll-out of BT's internal fibre portfolio, he organised a comprehensive trial of the current OTIAN® equipment used to support fibre in the access network, and oversaw the integration of fibre into the 'Video-on-Demand' trial in Kesgrave and the 'Interactive TV' trial in Colchester and Ipswich.

